THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 3: **Pair of Linear Equations in Two Variables** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $12x + 5y = 70$ and $7x + 8y = 51$, then $x + y =$ 1

(a) 2 **(b)** 5 **(c)** 10 **(d)** 7

2. The area (in sq. units) of the triangle formed by the line $4x + 5y = 20$ and the coordinate axes is: 1

(a) 20 **(b)** $\frac{5}{2}$ **(c)** 10 **(d)** 200

3. If $x + y = 12$ and $xy = 35$, then $x^2 + y^2 =$ 1

(a) 74 **(b)** 109 **(c)** 144 **(d)** 214

4. If $9x + 5y = 66$ and $5x + 9y = 58$, then $x - y =$ 1

(a) 2 **(b)** 8 **(c)** 31 **(d)** $\frac{4}{7}$

5. The pair $kx + 6y = k - 6$ and $24x + ky = k$ has no solution when $k =$ 1

(a) 12 **(b)** 6 **(c)** -12 **(d)** 24

6. The pair $kx + 10y = k - 10$ and $40x + ky = k$ has infinitely many solutions when $k =$ 1

(a) 20 **(b)** 40 **(c)** -20 **(d)** 10

7. If $15x + 4y = 66$ and $7x + 6y = 68$, then $x + y =$ 1

(a) 11 **(b)** 2 **(c)** 9 **(d)** 18

8. The area (in sq. units) of the triangle formed by the line $6x + 3y = 18$ and the coordinate axes is: 1

(a) $\frac{3}{2}$ **(b)** 162 **(c)** 18 **(d)** 9

9. **Assertion (A):** The pair $kx + 3y = 3$ and $12x + ky = 6$ has no solution when $k = -6$. 1

Reason (R): A pair has no solution when $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** If $2x + y = 5$ and $4x + 2y = k$ have infinitely many solutions, then $k = 10$. 1

Reason (R): A pair of linear equations in two variables can have exactly two solutions.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the points where the line $6x + 4y = 24$ meets the coordinate axes and the area of the triangle so formed. 2

12. Find k so that the pair $kx + 5y = k - 5$, $20x + ky = k$ has no solution. 2

13. Solve: $9x + 5y = 66$ and $5x + 9y = 58$. 2

14. Find the points where the line $5x + 3y = 15$ meets the coordinate axes and the area of the triangle so formed. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve: $\frac{3}{x} + \frac{2}{y} = \frac{19}{10}$ and $\frac{4}{x} + \frac{5}{y} = 3$. 3

16. Find a and b for which the pair $5x + y = 9$ and $(a + b)x + (a - b)y = 18$ has infinitely many solutions. 3

17. Find the area of the triangle formed by the lines $x + 3y = 9$, $2x + y = 8$ and the x-axis. 3

OR

Solve: $\frac{5}{x} + \frac{2}{y} = \frac{3}{2}$ and $\frac{3}{x} + \frac{4}{y} = \frac{11}{6}$. 3

18. Find a and b for which the pair $3x + y = 2$ and $(a + b)x + (a - b)y = 8$ has infinitely many solutions. 3

OR

Find the area of the triangle formed by the lines $2x + 3y = 12$, $x + y = 5$ and the x-axis. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Kavita thinks of a two-digit number. The sum of its digits is 9, and the number exceeds the number obtained by reversing its digits by 27. Let the tens digit be x and the units digit be y. 5

(i) Write the equation for the sum of digits.

(ii) Write and simplify the equation for the difference.

(iii) Find the number.

(iv) Show that no two-digit number has digit sum 9 and exceeds its reversal by 36.

20. A man travels 540 km partly by train and partly by bus. If he travels 270 km by train and the rest by bus, he takes $\frac{21}{2}$ hours; if he travels 360 km by train and the rest by bus, he takes 11 hours. Find the speeds of the train and the bus. 5

OR

Draw the graphs of $3x + y = 13$ and $x - 2y = 5$. Find the area of the triangle formed by these lines and the y-axis. 5

END OF QUESTION PAPER — Total: 40 Marks